

LESSON 2-5 • SESSION 1 • PRACTICE • FORM A**Range and IQR**

Learning target – I can find the range and the interquartile range of a data set. · 6.SP.B.5c

REMEMBER

- **Range** = greatest – least. It uses only the two extreme values.
- **IQR** = $Q3 - Q1$. It is the spread of the **middle half** of the data.
- Changing only the greatest value changes the range, **not** the IQR. Missing max = min + range.

WORKED EXAMPLE

Hikes (km): 5, 12, 8, 20, 15, 10, 18.

Step 1 Order: 5, 8, 10, 12, 15, 18, 20.**Step 2** Range = $20 - 5 = 15$ km.**Step 3** $Q1 = 8$, $Q3 = 18$, so $IQR = 18 - 8 = 10$ km.**Answer** Range 15 km; IQR 10 km**WORDS** range (*rango*) · interquartile range (IQR) (*rango intercuartil*) · variability (*variabilidad*)**PRACTICE**

show your work · label units

- 1. Camp Hikes.**
- Seven hikes covered 5, 12, 8, 20, 15, 10, 18 km.

Hint Order first. Range uses the ends; IQR uses $Q1$ and $Q3$.

- a. Range: ____ b. $Q1$: ____ c. $Q3$: ____ d. IQR: ____

- 2. Bake Sale.**
- Seven sales earned 48, 62, 55, 70, 51, 64, 59 dollars.

- a. Range: ____

- b. IQR: ____

- 3. Chorus Rehearsals.**
- Seven lasted 55, 72, 61, 88, 66, 79, 70 minutes.

Hint Leave the median out of both halves.

- a. Range: ____

- b. IQR: ____

- 4. Swapped Value.**
- A data set is 12, 14, 18, 22, 24, 29, 33. Someone replaces 33 with 95. Priscilla says both the range and the IQR grow.

Hint Only the greatest value changed.

- a. Range before; after: ____

- b. IQR before; after: ____

- c. Is she right? Explain.

- 5. Missing Maximum.**
- Seven values have min 7, range 38,
- $Q1 = 15$
- and
- $Q3 = 32$
- .

Hint Add the range to the minimum.

- a. Maximum: ____ b. IQR: ____

6. **Same Range?** Set C: 20, 24, 28, 32, 36, 40, 44. Set D: 20, 22, 23, 32, 41, 42, 44.

a. Range of C: ____ b. IQR of C: ____ c. Range of D: ____ d. IQR of D: ____

7. **Which Spread?** Hourly wages: min \$11, Q1 \$14, median \$16, Q3 \$19, max \$62.

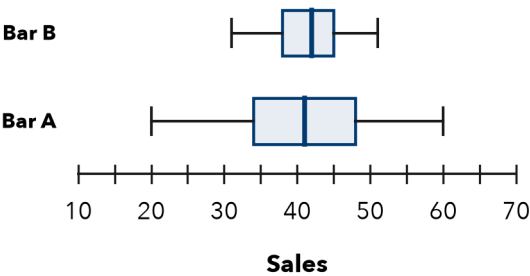
a. Range: ____ b. IQR: ____ c. Better for typical spread: ____

8. **Two Snack Bars.** The box plots show daily sales.

a. Range and IQR of each bar:

b. Which bar has the steadier middle half? _____

c. Why could the ranges mislead if one day had unusual weather?



FIND THE ERROR

explain the mistake, then fix it

9. **Tobias’s Work.** For 6, 10, 13, 19, 24, 30, 36, Tobias wrote: “The IQR is 30, the greatest value minus the least.” Explain the mistake and find the IQR.

TEST PRACTICE

choose the best answer

10. The range of a data set is found by –
- A Q3 – Q1 B greatest – least
- C adding all values D finding the middle
11. For min 18, Q1 25, median 30, Q3 39, max 52, the IQR is –
- A 34 B 14 C 9 D 21

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

17 M	15 E	51 E	27 Z	45 E	31 J	19 Q	10 X	18 R	13 T
#1a		#1d	#2b	#3b	#5a	#5b	#7a		

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____